Exercise 5.2 SOLUTION

Q1. Resolve into Partial Fractions $\frac{2x^2+3x+3}{(x+1)(x^2+1)}$

Sol: Suppose

$$\frac{2x^2+3x+3}{(x+1)(x^2+1)}=\frac{A}{x+1}+\frac{Bx+C}{x^2+1} \quad \text{_____ (i)}$$

Multiply by both sides $(x+1)(x^2+1)$, we get

$$2x^2+3x+3=A(x^2+1)+(Bx+C)(x+1) \quad \text{_____ (ii)}$$

Put $x+1=0 \Rightarrow x=-1$ in II

$$2(-1)^2+3(-1)+3=A\left((-1)^2+1\right)+\left(B(-1)+C\right)(-1+1)$$

$$2-3+3=A(1+1)+0 \quad =2=2A \Rightarrow \boxed{A=1}$$

Rearrange II

$$2x^2+3x+3=Ax^2+A+Bx^2+Bx+Cx+C$$

$$2x^2+3x+3=(A+B)x^2+(B+C)x+(A+C)x^0$$

Computing co-efficient

x^2; $A+B=2$ _____ (iii)

Put $A=1$ in equation (iii)

$1+B=2 \Rightarrow B=2-1 \Rightarrow B=1$

x; $B+C=3$ _____ (iv)

Put value $B=1$ in equation (iv)

$1+C=3 \quad =C=3-1 \Rightarrow 2=C$

Put values A,B and C in equation (i)

Hence $\frac{2x^2+3x+3}{(x+1)(x^2+1)}=\frac{1}{x+1}+\frac{x+2}{x^2+1}$

نوٹ

اگر denominator میں ایسا factor ہو جس میں x^2 ہو جیسے x^2+1 تو partial fraction بناتے وقت اس کے اوپر $(Ax+B)$ آئے گا لیکن شرط ہے کہ x^2+1 کو ختم کر کے x یعنی linear میں تبدیل نہ کیا جا سکتا ہو۔

Q2. $\frac{2x+1}{(x-2)(x^2+3x+5)}$

Sol: Suppose $\frac{2x+1}{(x-2)(x^2+3x+5)}=\frac{A}{x-2}+\frac{Bx+C}{x^2+3x+5}$ _____ (i)

Multiply by $(x-2)(x^2+3x+5)$ we get

$$2x+1=A(x^2+3x+5)+(Bx+C)(x-2) \quad \text{_____ (ii)}$$

Put $x-2=0 \Rightarrow x=2$ in equation (ii)

$$2(2)+1=A\left((2)^2+3(2)+5\right)+\left(B(2)+C\right)(2-2)$$

$$4+1=A(4+6+5)+0$$

$$5=\quad 15A \Rightarrow \frac{5}{15}=A \quad \Rightarrow A=\frac{1}{3}$$

Rearrange equation II

$$2x+1=Ax^2+3Ax+5A+Bx^2-2Bx+Cx-2C$$

$$2x+1=(A+B)x^2+(3A-2B+C)x+(5A-2C)x^0$$

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Comparing co-efficient

x^2; $A+B=0$ _____ (iii)

x; $3A-2B+C=2$ _____ (iv)

x; $5A-2C=1$ _____ (v)

Put value $A=\frac{1}{3}$ in equation (iii)

$$\frac{1}{3}+B=0 \Rightarrow B=\frac{-1}{3}$$

Put value $A=\frac{1}{3}$ in equation (v)

$$5\left(\frac{1}{3}\right)-2C=1 \Rightarrow \frac{5}{3}-2C=1$$

$$-2C=\frac{1-5}{3} \Rightarrow -2C=\frac{-2}{3} = C=\frac{1}{3}$$

Put values A, B and C in equation (i)

Hence $$\frac{2x+1}{(x-2)(x^2+3x+5)}=\frac{1}{3(x-2)}+\frac{\frac{-x}{3}+\frac{1}{3}}{(x^2+3x+5)}=\frac{1}{3(x-2)}-\frac{x-1}{3(x^2+3x+5)}$$

Q3. $$\frac{2x+32}{(x-2)(x^2+2)^2}$$

Sol: Suppose

$$\frac{2x+32}{(x-2)(x^2+2)^2}=\frac{A}{(x-2)}+\frac{Bx+C}{(x^2+2)}+\frac{Dx+E}{(x^2+2)^2} \quad \text{_____ (i)}$$

Multiply by $(x-2)(x^2+2)^2$ we get

$$2x+32=A(x^2+2)^2+(Bx+C)(x-2)(x^2+2)+(Dx+E)(x-2) \quad \text{_____ (ii)}$$

Put $x-2=0 \Rightarrow x=2$ in equation (ii)

$$2(2)+32=A((2)^2+2)^2+0+0$$

$$4+32=A(4+2)^2=36=36A \Rightarrow A=1$$

Rearrange equation (ii)

$$2x+32=A(x^4+4x^2+4)+(Bx+C)(x^3+2x-2x^2-4)+(Dx+E)(x-2)$$

$$2x+32=Ax^4+4Ax^2+4A+Bx^4+2Bx^2-2Bx^3-4Bx+Cx^3+2Cx-2Cx^2-4C+Dx^2-2Dx+Ex-2E$$

$$2x+32=(A+B)x^4+(-2B+C)x^3$$

$$(4A+2B-2C+D)x^2+(-4B+2C-2D+E)x+(4A-4C-2E)x^0$$

Comparing co-efficient

x^4; $A+B=0$ _____ (iii)

x^3; $-2B+C=0$ _____ (iv)

x^2; $4A+2B-2C+D=0$ _____ (v)

x; $-4B+2C-2D+E=2$ _____ (vi)

x; $4A-4C-2E=32$ _____ (vii)

Put $A=1$ in equation (iii)

$1+B=0 \Rightarrow B=-1$

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Put value $B=-1$ in equation (iv)

$-2(-1)+C=0 \Rightarrow 2+C=0 \Rightarrow C=-2$

Put A,B and C in equation (v)

$4(1)+2(-1)-2(-2)+D=0$

$4-2+4+D=0 \Rightarrow D=-6$

Put A , and C in equation (vii)

$4(1)-4(-2)-2E=32$

$4+8-2E=32 \quad =-2E=32-12$

$-2E=20 \Rightarrow E=-10$

Put values A,B,C,D and E in equation (i)

$$\frac{2x+32}{(x-2)(x^2+2)^2}=\frac{1}{x-2}+\frac{-x-2}{(x^2+2)}+\frac{-6x-10}{(x^2+2)^2}$$

Hence $$\frac{2x+32}{(x-2)(x^2+2)^2}=\frac{1}{x-2}-\frac{x+2}{(x^2+2)}-\frac{2(3x+5)}{(x^2+2)^2}$$

Q4. $\dfrac{3x^2+3}{x^3+1}$

Sol: Suppose

$$\frac{3x^2+3}{x^3+1}=\frac{3x^2+3}{(x+1)(x^2-x+1)}=\frac{A}{x+1}+\frac{Bx+C}{x^2-x+1} \quad \text{_____ (i)} \qquad \left\{a^3+b^3=(a+b)(a^2-ab+b^2)\right\}$$

Multiply by $(x+1)(x^2-x+1)$ we get

$3x^2+3=A(x^2-x+1)+(Bx+C)(x+1)$ _____(ii)

Put $x+1=0 \Rightarrow x=-1$ in equation (ii)

$3(-1)^2+3=A((-1)^2-(-1)+1)+(B(-1)+C)(-1+1)$

$3+3=A(1+1+1)+0 \quad \Rightarrow 6=3A \Rightarrow A=2$

Re-arrange equation ii

$3x^2+3=Ax^2-Ax+A+Bx^2+Bx+Cx+C$

$3x^2+3=(A+B)x^2+(-A+B+C)x+(A+C)x^0$

Comparing co-efficient

$x^2;\ A+B=3$ _____ (iii)

$x;\ -A+B+C=0$ _____ (iv)

$x;\ A+C=3$ _____ (v)

Put $A=2$ in equation (iii)

$2+B=3 \Rightarrow B=3-2 \Rightarrow B=1$

Put values A in equation (v)

$2+C=3 \quad =C=3-2 \Rightarrow C=1$

Put values A,B and C in equation (i)

Hence $$\frac{3x^2+3}{x^3+1}=\frac{2}{x+1}+\frac{x+1}{x^2-x+1}$$

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Q5. $\dfrac{x^4}{x^4+2x^2+1}$

Sol: $\dfrac{x^4}{x^4+2x^2+1}$ (Improper Fraction)

$$= 1 + \frac{-2x^2-1}{x^4+2x^2+1} = 1 - \frac{2x^2+1}{(x^2+1)^2} \quad \text{——— (i)}$$

$$x^4+2x^2+1 \overline{)\, x^4 \pm 2x^2 \pm 1} \;\; 1$$

$$-2x^2-1$$

Now we take $\dfrac{2x^2+1}{(x^2+1)^2}$

Suppose $\dfrac{2x^2+1}{(x^2+1)^2} = \dfrac{Ax+B}{(x^2+1)} + \dfrac{Cx+D}{(x^2+1)^2}$ ——— (ii)

Multiply by $(x^2+1)^2$, we get

$2x^2+1 = (Ax+B)(x^2+1) + (Cx+D)$ ——— (iii)

Re-arrange equation (iii)

$2x^2+1 = Ax^3 + Ax + Bx^2 + B + Cx + D$

Comparing co-efficients

x^3; $A = 0$

x^2; $B = 2$

x; $A + C = 0$ ——— (iv)

x^0; $B + D = 1$ ——— (v)

Put $A = 0$ in equation (iv)

$A + C = 0 \Rightarrow 0 + C = 0 \Rightarrow C = 0$

Put $B = 2$ in equation (v)

$2 + D = 1 \Rightarrow D = 1 - 2 \Rightarrow D = -1$

Put values A,B,C and D in equation (ii)

$\dfrac{2x^2+1}{(x^2+1)^2} = \dfrac{2}{x^2+1} - \dfrac{1}{(x^2+1)^2}$ put in equation (i)

Hence $\dfrac{x^4}{x^4+2x^2+1} = 1 - \dfrac{2}{x^2+1} + \dfrac{1}{(x^2+1)^2}$

Q6. $\dfrac{6x^4+40x^2}{(4-x^2)(x^2+4)^2}$

Sol: Suppose

$$\frac{6x^4+40x^2}{(4-x^2)(x^2+4)^2} = \frac{6x^4+40x^2}{(2-x)(2+x)(x^2+4)^2}$$

$$\frac{6x^4+40x^2}{(2-x)(2+x)(x^2+4)^2} = \frac{A}{(2-x)} + \frac{B}{(2+x)} + \frac{Cx+D}{(x^2+4)} + \frac{Ex+F}{(x^2+4)^2} \quad \text{——— (i)}$$

Multiply by $(2-x)(2+x)(x^2+4)^2$ we get

$6x^4+40x^2 = A(2+x)(x^2+4)^2 + B(2-x)(x^2+4)^2 + (Cx+D)(2-x)(2+x)(x^2+4) + (Ex+F)(2-x)(2+x)$ ——— (ii)

Put $2 - x = 0 \Rightarrow 2 = x$ in equation (ii)

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$6(2)^4+40(2)^2=A(2+2)\left((2)^2+4\right)^2+B(2-2)\left((2)^2+4\right)^2+(C(2)+D)(2-2)(2+2)\left((2)^2+4\right)+(E(2)+F)(2-2)(2+2)$

$96+160=A(4)(64)+0+0+0 \Rightarrow 256=256A \quad \Rightarrow A=1$

Put $2+x=0 \quad \Rightarrow \quad x=-2$ in equation (ii)

$6(-2)^4+40(-2)^2=0+B(+2+2)\left((-2)^2+4\right)^2+0+0$

$96+160=B(4)(64) \Rightarrow 256=256\,B \quad \Rightarrow B=1$

Re-arrange equation ii

$6x^4+40x^2=A(2+x)\left(x^4+16+8x^2\right)+B(2-x)\left(x^4+16+8x^2\right)+(Cx+D)\left(4-x^2\right)\left(4+x^2\right)+(Ex+F)\left(4-x^2\right)$

$6x^4+40x^2=A\left(2x^4+32+16x^2+x^5+16x+8x^3\right)+B\left(2x^4+32+16x^2-x^5-16x-8x^3\right)+(Cx+D)\left(16-x^4\right)+(Ex+F)\left(4-x^2\right)$

$6x^4+40x^2=A\left(2x^4+32+16x^2+x^5+16x+8x^3\right)+B\left(2x^4+32+16x^2-x^5-16x-8x^3\right)+\left(16Cx-Cx^5+D16-Dx^4\right)+\left(4Ex-Ex^3+4F-Fx^2\right)$

Comparing co-efficients

x^5; $A-B-C=0$ _____ (iii)

x^4; $2A+2B-D=6$ _____ (iv)

x^3; $8A-8B-E=0$ _____ (v)

x^2; $16A+16B-F=40$ _____ (vi)

x; $16A-16B+16C+4E=0$

$4A-4B+4C+E=0$ _____ (vii)

x^0; $32A+32B+16D+4F=0$

$8A+8B+4D+F=0$ _____ (viii)

Put values A and B in Equation (iii)

$A-B-C=0 \Rightarrow 1-1-C=0 \quad \Rightarrow \quad C=0$

Put value A and B in Equation (iv)

$2(1)+2(1)-D=6 \Rightarrow 4-D=6 \Rightarrow -D=6-4 \Rightarrow D=-2$

Put value A and B in Equation (v)

$8(1)-8(1)-E=0 \Rightarrow E=0$

Put value A and B in equation (vi)

$16(1)+16(1)-F=40 \quad =-F=40-32 \Rightarrow -F=8 \Rightarrow F=-8$

Equation (i) Become

Hence $\dfrac{6x^4+40x^2}{(2-x)(2+x)\left(x^2+4\right)^2}=\dfrac{1}{2-x}+\dfrac{1}{2+x}-\dfrac{2}{\left(x^2+4\right)}-\dfrac{8}{\left(x^2+4\right)^2}$

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